

Engineer with practical background in cybersecurity and data science. I have built and deployed end-to-end solutions across AI, ML and security. Seeking Werkstudent opportunities where I can contribute while continuing to grow as an engineer.

SKILLS

Languages	Python, JavaScript, Java, C
Backend	FastAPI, Flask, Express.js, Node.js
Frontend	React, HTML5, Tailwind CSS
AI & ML	NumPy, Pandas, Scikit-learn, LLMs, OpenAI API, RAG, LangChain
Databases	MongoDB, PostgreSQL, MySQL, ChromaDB
Tools	Git, Docker, Postman, Nmap, Kali Linux

PROJECTS

Credit Card Customer Churn Analytics Platform

Live Demo | [GitHub](#)

June 2026

Scikit-Learn, XGBoost, FastApi, PostgreSQL

- Analyzed **10,127 customer records**, uncovering a **16.07% churn rate** through interactive dashboards and customer segmentation.
- Developed a churn prediction model achieving **ROC-AUC 0.993, F2-score 0.919** and **Average Precision 0.968**.
- Identified **1,354 high-risk customers**, **\$609K revenue at risk**, and **\$152K** potential annual savings through **25%** retention scenario simulation.

RAG Study Application: Study Smart

Live Demo | [GitHub](#)

May 2026

RAG, LangChain, HuggingFace, LLM APIs

- Engineered a RAG pipeline using **900 character text chunking** and **18%** (**160-character**) overlap for optimal contextual continuity.
- Built dual-stage retrieval system extracting **12** candidate chunks across **1536-D** vector space, using a Cross-Encoder to rerank the top **4** results.
- Architected dockerized full-stack app for PDF uploads, document chat, reranked chunk score visualization, summarization, and MCQ quiz generation.

RecipeHub

Live Demo | [GitHub](#)

April 2025

MongoDB, Express.js, React.js, Node.js

- Developed a **dockerized MERN platform** featuring live recipe search, pantry matcher, dynamic meal planner, and a user-favourite bookmarking system.
- The feature **pantry matcher** processes existing kitchen ingredients to calculate an exact % match scoring system for optimized dish discovery.
- Meal planner** generates dynamic **3-to-7** day schedules while handling real-time requests across **4** dietary, **3** allergy, and **8** cuisine filters.

EDUCATION

Master of Science in Data Science

Hamburg University of Technology (TUHH)

Oct 2026

Bachelor of Engineering in Computer Science and Engineering (Cyber Security)

Visvesvaraya Technological University (VTU)

2022 – 2026

CGPA: 9.2/10

PUBLICATIONS

From Sky to Spectrum: Unveiling the Potential of 6G for UAV Communication

Published in: IEEE Xplore via the International Conference on Computing for Sustainability and Intelligent Future (COMP-SIF): Highlights the convergence of Machine Learning, security, and trajectory optimization in UAVs using 6G.

April 2025